OpenADMET CYP Challenge — Structure Prediction Track
Model Description Report

Team / leaderboard alias: Cheminfo

Submission: 20 CYP3A4 protein–ligand complexes, predicted from ligand SMILES alone

Format: one archive of 20 flat `{Molecule_Name}.pdb` files; chain A protein, chain B haem, chain C ligand as residue `LIG`

Evaluation metrics: LDDT-PLI (primary), BiSyRMSD (secondary), LDDT-LP (protein), with PoseBusters as a validity gate

1. Task and scoring

Twenty blinded ligands — fifteen from the training set, two from the blinded 750-compound test set, three additional compounds — are to be placed in CYP3A4, with twenty novel UCSF cryo-EM structures as the blinded reference. No receptor structure is provided; the complex must be produced from sequence and SMILES.

Two properties of the scoring shaped the whole approach:

1. **PoseBusters is a zero-score gate, not a penalty.** A structure that fails validity scores LDDT-PLI 0 for that ligand, not a reduced score. A structure that the scorer cannot map — duplicate `(chain, resid, icode)` keys, duplicate atom names within a residue, duplicate atom serials — scores 0 *and* takes a 20 Å BiSyRMSD penalty. One invalid pose therefore costs 5% of the primary metric outright.
2. **The objective separates into twenty independent arg-maxes.** LDDT-PLI is a per-ligand mean and the validity zero is per-ligand, so there is no combinatorial pose-selection problem — each ligand can be optimised alone.

This was measured, not assumed. Two submissions differing in exactly five files invert to give 0.322 mean LDDT-PLI for the fifteen shared poses and 0.754 for the five replaced ones — a spread only consistent with invalid structures scoring zero.

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2. Pose generation

Two independent pose sources were built, and the submitted archives are hybrids of them.

2.1 Boltz-2 co-folding

Setting	Value
Model	Boltz-2, version 2.2.1
Protein specification	UniProt P08684 canonical sequence, 503 residues — sequence and MSA only, **no structural template**
MSA	14,302 sequences, computed once with the ColabFold MSA server and shared across every ligand and seed
Cofactor	Haem B supplied as CCD entry `HEM`, so atom names match the references
Pocket restraint	24 distal (substrate-side) residue contacts, soft (`force: false`), 8 Å. Cys442 deliberately excluded as proximal-face
Sampling	5 diffusion samples per seed × 3 seeds × 20 ligands = **300 complexes**

Steering potentials	Enabled, which keeps bond lengths and clashes inside PoseBusters tolerances

2.2 gnina ensemble docking

The 300 Boltz complexes were stripped to protein plus haem — each one's own ligand discarded — and used as a receptor ensemble. The search box is **identical in every frame**: a 24 Å cube centred 7.0 Å along the haem distal normal from the iron, rather than auto-boxed, so scores are comparable across frames. The distal face was correctly identified in 300 of 300 frames and the box encloses each frame's native ligand in 300 of 300.

Ligands were protonated for pH 7.4 using OpenBabel's pH model, SMARTS-audited per ionisable group; all twenty are neutral, confirmed two independent ways. `dimorphite-dl` was evaluated and rejected — it returns protonated amide, urea and sulfonamide nitrogens and a net +3 charge on one ligand at pH 7.4.

Twelve ligands carrying a coordinating nitrogen were docked with a covalent pin to the haem iron; the remaining eight were free-docked. Both used `--cnn_scoring rescore --num_modes 9 --exhaustiveness 8 --seed 42`. In total 6,000 docking runs produced 50,509 poses with no failures, of which 28,909 pinned poses were retained.

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3. Iron coordination — the central problem

The diagnosis that drove the method: **Boltz systematically fails to coordinate the haem iron.** Ten of the twenty ligands carry a coordinating nitrogen and sit at an iron distance of 2.83–2.87 Å — a spike, not a distribution, with all ten inside 0.04 Å of each other — where a real azole or pyridine to Fe(III) bond is 2.0–2.2 Å. The organisers' own baseline places the same ten at 2.00–2.14 Å. A SMARTS audit separates the set perfectly: ten close with a coordinating nitrogen, zero close without one.

A forced Fe–N distance restraint was written into the Boltz input at a 2.2 Å target and **did not achieve it**. Across 150 constrained diffusion samples every single one landed in 2.83–2.88 Å. The constraint is nevertheless not a no-op — a ligand starting at 5.28 Å was pulled to 2.83–2.87 Å, a 2.4 Å move — so 2.83 Å is best understood as a **learned floor** in the model, and Boltz exposes no constraint-strength parameter with which to push past it.

The route that worked was a different sampler. gnina's covalent docking mode pins a named ligand atom to a fixed point: the iron position as receptor atom, and an axial site 2.10 Å along the distal haem normal as the target position. This achieved **exactly 2.100 Å on 100% of all 28,909 pinned poses**, verified by re-measuring written coordinates rather than trusting exit codes.

> **The pin mechanism has a trap worth documenting.** gnina cannot pin by atom index — it pins the first atom of the first SMARTS match. An unrooted pattern therefore pins a *carbon* to the iron. Each pattern must be rooted at the intended atom, and even a single-match generic pattern can select the wrong one: on one ligand a generic imidazole pattern picked the non-coordinating triazole nitrogen. The fix was a per-ligand smallest rooted environment SMARTS selecting exactly the intended atom.

A second, finer defect was then identified: the pin fixes distance and axial tilt by construction but **not orientation**. Two ligands have the iron roughly 30° out of the coordinating ring plane — toward the π face rather than along the nitrogen lone pair — which is not σ coordination. Distance alone is not coordination, and PoseBusters does not detect this: it passes all thirteen distorted-ring Boltz poses. It is a validity gate with generous tolerances, never a quality measure.

4. Pose selection

Three selection rules were used, for different pose sources.

Rule	Definition	Applied to
Boltz confidence	Per-ligand arg-max of `ligand_iptm`	First submission only
Medoid	Per ligand, the pose with lowest mean RMSD to all other poses — the consensus. Poses superposed on the haem first, atom correspondence exact by name	Boltz-only archives
Rank-1 CNNaffinity	gnina's CNN affinity head, with the exported picks forced to come from different protein conformations	All docked poses in the submitted archives

The medoid replaced confidence ranking because confidence cannot discriminate between genuinely different poses: within-ligand `ligand_iptm` spread is a median of 0.0068 while within-ligand geometric spread is 0.33–4.33 Å mean RMSD and up to 6.77 Å pairwise. On one seed the medoid changes 15 of 20 picks for a mean confidence cost of 0.0042.

> **Two selection proxies that were tested and refuted.** First, ranking poses on RMSD to the organisers' baseline structures: it ranks the submission that gained +0.0913 and nineteen places as the *worst* candidate, and is not monotone with LDDT-PLI across pose sources. Second, selecting on docking score is close to adversarial for this target — Vina-family scoring treats iron as a generic atom, so a correct 2.1 Å coordination reads as a clash and the affinity head steers away from it. The gate that *did* hold is the simplest: the count of ligands with an Fe–N contact at or below 2.2 Å. An archive with eleven coordinating contacts read 0.4300; one with four read 0.4111.

5. PoseBusters repair

Rather than guessing thresholds, PoseBusters itself is used as the oracle. For each failing ligand the pipeline walks down the CNNaffinity-ranked list of poses *in distinct receptors*, exports the first twelve candidates, runs the real gate over the batch, and takes the highest-scoring candidate that passes. A pinned pose is eligible only if its pin actually fired (2.10 ± 0.25 Å), so a repair can never be a pose that quietly lost axial coordination.

A cheap distance screen was insufficient because PoseBusters scales its cutoff by the two atoms' van der Waals radii rather than using a flat distance, and it checks the porphyrin as an organic cofactor, which a blanket haem exemption lets through. Reaching full validity cost very little in coordination: mean ligand–iron distance moved from 2.895 to 2.970 Å.

Archive	PoseBusters
Organisers' Boltz-2 baseline	19 / 20
All gnina rank-1	15 / 20
Rank-2 substitution	16 / 20
Targeted repair	17 / 20
clean20 and all later archives	**20 / 20**

6. Submission history

Date (UTC)	Archive	Method	PB	LDDT-PLI	Rank

2026-10-13	seed1	Boltz, 1 seed, confidence pick	20/20	0.4106	13 / 29
2026-10-15	run2	Boltz, 3 constrained seeds, medoid pick	20/20	0.4137	20 / 36
2026-10-26	dock	All gnina rank-1	15/20	0.3387	32 / 41
2026-10-27	**clean20**	gnina rank-1 + 5 repaired poses	**20/20**	**0.4300**	**13 / 41**
2026-10-28	hybridH	Boltz run2 + clean20's 5 repairs	20/20	0.4111	20 / 41
2026-10-28	hybridHC	clean20 chelators + Boltz non-chelators	20/20	0.4293	17 / 41
2026-11-02	geomfix	clean20 with 2 geometry re-picks	20/20	0.3926	28 / 48

The method is best read off three of these. Moving from Boltz poses to validity-repaired docked poses (`run2` → `clean20`) gained +0.0163 and was the single largest improvement; it is entirely attributable to iron coordination, since the Boltz poses in `run2` sit at the 2.83 Å floor and the docked poses at 2.10 Å. Replacing the docked non-chelator poses with Boltz ones (`hybridHC`) read null at 0.4293, which localises the gain to the coordinating ligands specifically rather than to pose quality in general. The final entry is a two-ligand geometry re-pick probe that regressed and does not represent the method; `clean20` remains the best configuration on this track.

> **Reproducibility note.** The archive checksum is not reproducible. A rebuild produces all twenty PDB files byte-identical, verified per member, but a different archive hash, because the format embeds modification times. Archives are therefore retained as artefacts rather than regenerated.

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7. Software and hardware

Tool	Version	Role
Boltz	2.2.1 (Boltz-2)	Co-folding of CYP3A4–ligand complexes
gnina	1.3.3	Ensemble docking and covalent iron pinning
PoseBusters	0.6.4	Validity gate (`dock` mode)
RDKit	2025.3.6 / 2025.9.6	Ligand handling, SMARTS, atom naming
gemmi	0.7.4 / 0.7.5	CIF and PDB input/output in the archive builder
MDAnalysis	2.9.0	Challenge structure validator
OpenBabel	—	pH 7.4 ligand protonation
ColabFold MSA server	—	Protein multiple sequence alignment

Boltz inference ran on an NVIDIA Tesla T4 (16 GB) cloud instance at one worker; free docking ran on a local RTX 4060 at 16 workers and roughly 25 runs per minute, and pinned docking, which costs about three times as much per run, on the same hardware. A 503-residue complex with haem is about 4,100 atoms and needs 6–10 GB of VRAM per worker.